

Refrigerated and Frozen Storage Temperature Control and Cold-Chain Verification

Distribution Center DC-2 — Cold Rooms A/B and Freezer F1

SOP Number	SOP-WH-021
Revision	2.2
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Document Owner	Warehouse Operations Dept.
Approved By	Distribution Quality Lead
Applicable Area(s)	DC-2 Cold Rooms A/B, Freezer F1
Classification	Internal - Controlled

1. Purpose

This procedure defines the requirements for monitoring, recording, and verifying the temperature of refrigerated and frozen product during receiving, storage, and shipping at Distribution Center DC-2. It is intended to maintain an unbroken cold chain and to provide a documented, repeatable basis for shipment hold and release decisions.

2. Scope

This SOP applies to all refrigerated and frozen finished goods received, stored, and shipped through DC-2 Cold Rooms A and B and Freezer F1. It covers receiving temperature checks, continuous storage monitoring, temperature excursions, and pre-shipment verification. It does not cover ambient dry-goods storage, which is addressed in SOP-WH-004.

3. Definitions

Cold Chain: The uninterrupted series of temperature-controlled storage and handling steps that keep a product within its required temperature range from receipt through shipment.

TOR (Time-Out-of-Refrigeration): The cumulative time a refrigerated product spends above 40 degrees F (4 degrees C) while outside controlled cold storage.

Temperature Excursion: Any recorded event in which a storage area's air temperature moves outside its validated setpoint range for more than 15 minutes.

Setpoint: The target air temperature a storage area is controlled to, distinct from the alarm limit at which corrective action begins.

Cold-Chain Release: The documented sign-off confirming a shipment met temperature requirements throughout storage and is cleared to ship.

Data Logger: A calibrated electronic device that records air or product temperature at fixed intervals.

4. Responsibilities

Role	Responsibilities
Receiving Clerk	Measures and records product temperature at receipt and completes the Receiving Temperature Log before product is moved to storage.
Cold Storage Operator	Monitors area temperatures, responds to alarms, and records corrective actions on the Excursion Log.
Warehouse QA Technician	Performs pre-shipment temperature verification and approves or rejects shipments for cold-chain release.
Shift Lead	Confirms the shipping schedule, resolves cold-dock conflicts, and co-signs the Cold-Chain Release Record.
Distribution Quality Lead	Owens this SOP, reviews excursion and deviation reports, and approves any temporary variance to stated parameters.

5. Materials and Equipment

- Calibrated data loggers: One per storage area, logging air temperature at 15-minute intervals
- Handheld probe thermometer: Calibrated; used for product core temperature checks
- Cold Rooms A and B: Validated setpoint 36 degrees F (2 degrees C), acceptable range 34-40 degrees F
- Freezer F1: Validated setpoint -10 degrees F, acceptable range -20 to 0 degrees F
- Cold-chain forms: Excursion Log, Receiving Temperature Log, and Cold-Chain Release Record

6. Procedure

6.1 Receiving Inspection

1. Receiving Clerk verifies that the carrier's transport temperature record accompanies the shipment.
2. Measure the product core temperature of at least three cases per pallet using the calibrated probe thermometer.
3. Refrigerated product must be at or below 40 degrees F to be accepted into Cold Rooms A/B; frozen product must be at or below 0 degrees F to be accepted into Freezer F1.
4. Reject and quarantine any refrigerated shipment with a product core temperature above 45 degrees F; place it on hold pending disposition.
5. Record all receiving temperatures on the Receiving Temperature Log before product is moved to storage.

6.2 Storage Monitoring

6. Each storage area is continuously monitored by a calibrated data logger recording air temperature every 15 minutes.

7. Cold Storage Operator reviews logger readings at the start of every shift and records the review on the Storage Monitoring Log.
8. Product core temperature is spot-checked and recorded every 4 hours for product staged on the cold dock.
9. The cumulative time-out-of-refrigeration (TOR) for any refrigerated product staged outside cold storage must not exceed 2 hours.

6.3 Temperature Excursions

10. An area temperature outside its validated range for more than 15 minutes is a Temperature Excursion and triggers an alarm.
11. Cold Storage Operator responds to the alarm, identifies the cause, and records the event and corrective action on the Excursion Log.
12. Product exposed to an excursion is placed on hold and evaluated for disposition before it may ship.

6.4 Pre-Shipment Verification

13. Warehouse QA Technician reviews the storage monitoring record for the shipment's storage period and confirms no unresolved excursion applies.
14. Measure and record the product core temperature of a minimum of five cases from the outbound shipment.
15. A refrigerated shipment passes when every measured case is at or below 40 degrees F; any case above 40 degrees F requires the shipment to be held and re-evaluated before release.
16. Warehouse QA Technician completes and signs the Cold-Chain Release Record; Shift Lead co-signs before the shipment may leave the dock.

7. Corrective Actions

If any verification or monitoring step fails, the following actions apply:

- Product affected by an excursion or a failed verification is placed on hold and evaluated for disposition before release.
- Two consecutive failed pre-shipment verifications on the same shipment trigger a documented deviation report and notification to the Distribution Quality Lead before a third attempt.
- Any shipment loaded before a passing verification is recorded is recalled from the dock and placed on hold.
- Repeated excursions (3 or more in a rolling 30-day period) in the same storage area trigger a root-cause investigation per SOP-WH-030.

8. Documentation and Record Retention

The Receiving Temperature Log, Storage Monitoring Log, Excursion Log, data logger exports, and Cold-Chain Release Records generated by this procedure are retained by Warehouse Operations for a minimum of 12 months from the date of creation.

9. Training Requirements

All personnel performing tasks under this SOP must complete cold-chain handling and temperature-monitoring training and demonstrate competency before working unsupervised. Refresher training is required annually and whenever this SOP is revised at a level that changes a stated parameter (temperature, time, or interval).

10. Related Documents

- SOP-WH-004 – Ambient Dry-Goods Storage
- SOP-WH-030 – Root Cause Investigation, Distribution
- SOP-SN-005 – Warehouse Pest Control Program
- SOP-QA-011 – Product Disposition and Hold Procedure
- Attachment A – Receiving Temperature Log (form)
- Attachment B – Storage Area Setpoint and Alarm Limits

11. Revision History

Rev	Date	Author	Description of Change
2.0	2024-05-02	Warehouse Ops	Initial consolidated cold-chain SOP for DC-2, replacing area work instructions.
2.1	2025-07-18	Warehouse Ops	Added continuous data-logger monitoring at 15-minute intervals and the Excursion Log.
2.2	2026-04-15	Warehouse Ops	Lowered the refrigerated receiving rejection threshold from 47 to 45 degrees F.